

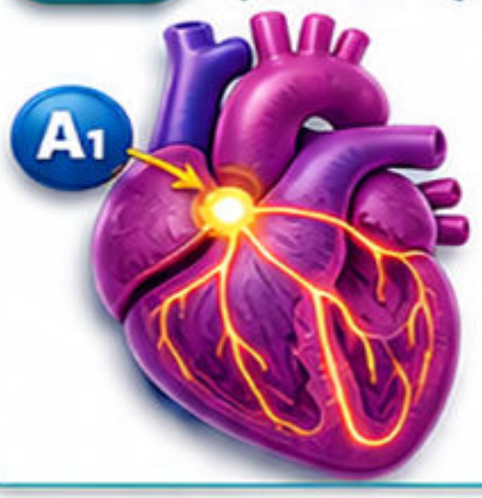
ADENOSINE

Pharma Talk

[ADENOCARD, ADENOSCAN]

CLASS: ANTIARRHYTHMIC, ENDOGENOUS NUCLEOSIDE

1 MECHANISM OF ACTION (MOA) & TARGET



- **TARGET:** A1 receptors in AV node
- **MOA:** Binds A1 → activates K^+ channels → hyperpolarization → slows AV conduction → terminates reentry
- **Analogy:** A temporary electrical reset switch for the heart

2 FORMS & STRENGTHS

- IV injection: 3 mg/mL



3 INDICATIONS (FDA-APPROVED & MAJOR OFF-LABEL)



FDA-APPROVED

- Conversion of paroxysmal SVT to normal sinus rhythm (adolescents and adults).
- Pharmacologic stress testing for detection of myocardial ischemia (adults).

MAJOR OFF-LABEL

- None established.

4 DOSAGE & ADMINISTRATION

A) ADULT DOSE



- Standard: 6 mg rapid IV push.
- Max: 12 mg per single dose.
- Frequency: May repeat once if needed.

B) PEDIATRIC DOSE



- Weight-Based: 0.1 mg/kg rapid IV push.
- If needed: 0.2 mg/kg rapid IV push.
- Max: 12 mg per dose.

C) ADMINISTRATION



- Route: Rapid IV push over 1–3 sec.
- Follow immediately with 20 mL NS flush.
- Elevate extremity.

D) ADJUSTMENTS



- Renal adjustment: None.
- Hepatic adjustment: None.

5 PHARMACOKINETICS & DYNAMICS (PK/PD)



- Onset: <10 sec
- Half-life ($t_{1/2}$): <10 sec
- Metabolism: Cellular uptake
- Renal excretion: Minimal
- Duration: Very brief

6 ADVERSE EFFECTS & TOXICITY



COMMON

- Facial flushing
- Chest discomfort
- Transient dyspnea



SEVERE / HIGH-RISK

- Prolonged asystole
- Severe bronchospasm
- High-degree AV block

7 CONTRAINDICATIONS & INTERACTIONS



ABSOLUTE CONTRAINDICATIONS

- 2nd-degree AV block
- 3rd-degree AV block
- Sick sinus syndrome
- Asthma (severe/reactive airway disease)



MAJOR DRUG INTERACTION

- Methylxanthines (e.g., caffeine, theophylline) → A1 receptor antagonism → decreased efficacy

8 SPECIAL POPULATIONS (PREGNANCY & LACTATION)



- Pregnancy: Use if clearly needed.
- Lactation: Caution advised.
- Geriatric: Standard dosing.

9 CLINICAL PEARLS



- Administer via proximal IV or central line for fastest delivery to the heart.
- Always follow with rapid NS flush and elevate the extremity.
- Avoid use in severe reactive airway disease.
- Dipyridamole blocks cellular uptake, drastically increasing adenosine levels; reduce dose by 75%.

10 GLOSSARY OF ABBREVIATIONS



AV = Atrioventricular



SVT = Supraventricular tachycardia



IV = Intravenous



NS = Normal saline